MAE3405 Aerospace Propulsion

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Topic 11-12 Problem Set Questions

An asterisk indicates that this problem is similar to a past exam question.

Question 29

A single stage rocket is being designed to launch a satellite. The launch angle of 80 degrees is used for a programmed burn time of 50 sec. The mass of the probe is 50kg, and over all structure mass 1500kg. The rocket motor has a combustion chamber pressure of 3.0MPa. If a convergent/divergent nozzle is used with throat area 0.1m^2, and it is designed to have an exit pressure equal to that found at a pressure altitude of 15km. Using the rocket fuel data table in the notes for following fuels:
(i) Kerosene/O_2 , and
(ii) H_2/O_2 ,

Calculate:

a) The exhaust exit velocity. [Ans: (i) 3061 m/s; (ii) 4423 m/s]
b) The propellant mass flow. [Ans: (i) 176 kg/s; (ii) 122 kg/s]
c) The thrust. [Ans:(i) 0.54 MN; (ii) 0.54MN]
d) The specific impulse. [Ans: (i) 312 s; (ii) 451 s]
e) Total propellant mass & volume.
[Ans: (i) 8.8t; 8240 m^3; (ii) 6095 kg; 23x10^3 m^3]
f) The velocity increment [Ans: (i) 5333 m/s; (ii) 6573 m/s]
g) Burnout height and coast height.
[Ans: (i) 89.9 km; 1966 km; (ii) 119 km; 3483km]

Question 30*

A single stage rocket is used to launch a satellite. A launch angle of 85º is used for a programmed burn time of 75 sec. The mass of the probe is 100kg, and mass of the rocket structure 3000kg. The rocket motor has a combustion chamber pressure of 3.0MPa and temperature of 2750ºC. A convergent/divergent nozzle is used with throat area 0.1m^2. The nozzle is designed to have an exit pressure equal to that found at SSL. Exhaust gas specific heat ratio is 1.15, and molecular weight 22 kg/kmol. Determine:

(i) the launch thrust at SSL

(ii) the motor burnout height assuming the nozzle exit velocity remains constant at its launch value.

[Ans: (i) 448kN, (ii) 88km]

Question 31

A rocket is being designed to launch a deep space probe. The total mass of the rocket is limited to 1000kg resulting in a structure stage fraction of 0.075 (constant for any number of stages). For the fuel used the rocket is expected to have an effective exhaust velocity of 4500m/s. Determine the variation in possible payload and total propellant mass if the rocket is made up of either (i) 1 or (ii) 2 stages.

[Ans: (i) 8 kg; 917kg; (ii) 45 kg; 864 kg]

Question 33*

A rocket motor is fuelled by a stoichiometric mix of hydrogen and oxygen. The fuel and oxidiser enter the combustion chamber at a temperature of -23ºC, then burn at a constant chamber pressure of 7.0 MPa. The efficiency of the combustor is 90%. The formation enthalpy of the water vapour produced from the combustion process is 241.827 MJ/kmol. The rocket uses a combustion tap-off cycle to power its turbopumps. For an altitude of 20,000m, determine:

(i) The thrust produced assuming a converging nozzle with exit diameter 0.5m and isentropic efficiency 95%.

(ii) Whether the thrust produced would be increased over (i) by replacing the nozzle with a converging-diverging nozzle with the same isentropic efficiency and throat area that expands the exhaust gases to atmospheric pressure for the given altitude.

[Ans: (i) 1.673 MN, (ii) 2.25 MN]

Question 33B

A rocket motor is fueled by a stoichiometric mix of kerosene ($C_{12}H_{26}$) and oxygen. The fuel and oxidiser enter the combustion chamber at a temperature of -23°C, then burn at a constant chamber pressure of 7.0 Mpa.

The rocket uses an open gas generator cycle, where 5% of the propellant is diverted to the preburner with the remainder burning in the main combustion chamber. The efficiency of the combustor and preburner are 90%.

Kerosene formation enthalpy is -291 MJ/kmol and C_P =2 kJ/kg-K. For other species take values of formation enthalpy and C_P at standard conditions as an estimate

For an altitude of 20,000m, determine the thrust and total propellant flow rate assuming a converging nozzle with exit diameter 0.5 m and isentropic efficiency 95%. [Ans: 1.66 MN, 630.3 kg/s]